

Question 1**(Begin a new page)****Marks**

- (a) Evaluate $\int_0^1 \frac{e^x}{1+e^x} dx$ 2
- (b) Find $\int \frac{1 - \sin x}{\cos^2 x} dx$ 2
- (c) (i) use the substitution $u = 1 - x^2$ to evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{x^3}{\sqrt{1-x^2}} dx$ 2
- (ii) Hence use integrating by parts to find $\int_0^{\frac{\sqrt{3}}{2}} 3x^2 \cos^{-1} x dx$. 2
- (d) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\pi/3} \frac{1}{5 - 4 \cos x} dx$, expressing the answer in simplest exact form. 3
- (e) (i) Find the real numbers A and B such that
$$\frac{x^2 + 3x}{(x^2 + 1)(x + 1)} = \frac{Ax + 1}{x^2 + 1} + \frac{B}{x + 1}$$
 2
- (ii) Hence find $\int \frac{x^2 + 3x}{(x^2 + 1)(x + 1)} dx$ 2

Question 2	(Begin a new page)	Marks
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(a)	Show that $(1 + i)^3 = 2(i - 1)$	1
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(b)	Show that the complex number $z = \frac{6 - 2i}{3 + 4i} - \frac{6}{5i}$ is real.	2
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(c)	Sketch the region in the Argand diagram where the inequalities	3
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$$|z - 2 + 2i| \leq 2 \text{ and } -\frac{\pi}{3} < \arg z \leq -\frac{\pi}{6}$$

holds simultaneously.

(d)	$z = x + iy$ is a complex number such that $ z - 3 - 4i = 2$.	
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(i)	Find the maximum value of $ z $.	2
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(ii)	Find the minimum value of $\arg z$.	2
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(e)	For any non-zero complex number z ,	
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(i)	show that $\arg\left(\frac{z}{\bar{z}}\right) = 2 \arg z$.	1
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(ii)	Let z be a complex number for which $z \neq 0$, $z \neq 1$ and	2
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$$\frac{z}{\bar{z}} = -\frac{z-1}{\overline{z-1}}$$

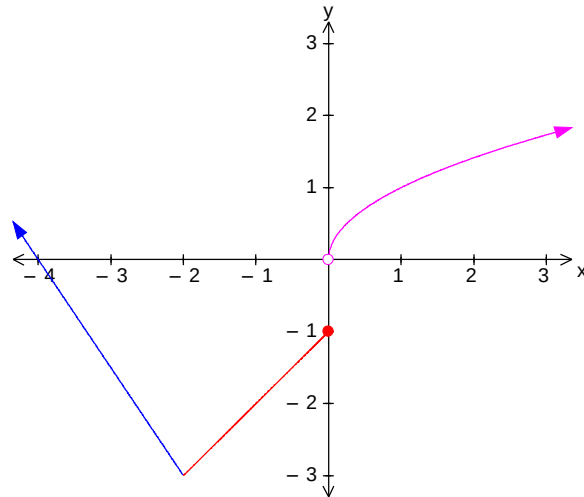
Show that $\arg z = \arg(z - 1) + \frac{\pi}{2}$ or $\arg z = \arg(z - 1) - \frac{\pi}{2}$

(iii)	Hence sketch the locus of all points z that satisfy	2
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$$\frac{z}{\bar{z}} = -\frac{z-1}{\overline{z-1}}$$

Question 3**(Begin a new page)****Marks**

- (a) The diagram below show the discontinuous function $y = f(x)$.



Draw large (half page), separate sketches of each of the following:

- | | |
|---|---|
| (i) $y = f(x - 1) $ | 2 |
| (ii) $y = \frac{1}{f(x)}$ | 2 |
| (iii) $y = \sqrt{-f(x)}$ | 2 |
| (iv) $y = \ln(f(x))$ | 2 |
| (v) $y^2 = f(x)$ | 2 |
| (b) It is given that $3 - i$ is a root of $P(z) = z^3 + kz + 60$, where k is a real number. | |
| (i) State why $3 + i$ is also a root of $P(z)$. | 1 |
| (ii) Solve the equation $P(z) = 0$ | 1 |
| (iii) Hence determine the value of k . | 1 |
| (c) For $z = r(\cos \theta + i \sin \theta)$, find r and the smallest value of θ which satisfies the equation $2z^3 = 9 + 3\sqrt{3}i$ | 2 |

Question 4

(Begin a new page)

Marks

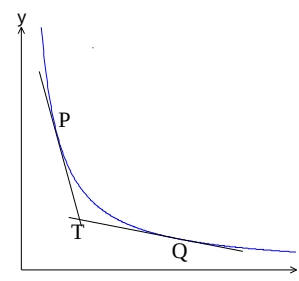
- (a) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $a > b > 0$ has eccentricity e .

(i) Show that the line through the focus $F(ae, 0)$ that is perpendicular to the asymptote $y = \frac{bx}{a}$ has equation $ax + by - a^2e = 0$ 1

(ii) Show that this line meets the asymptote at a point on the corresponding directrix. 2

- (b) Find the equation of the normal to the curve $x^4 + 3xy - 2y^2 + 13 = 0$ at the point $(-1, 2)$. 3

- (c) The point $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ $p \neq q$ lies on the same branch of the hyperbola $xy = c^2$. The tangents at P and Q meet at point T .

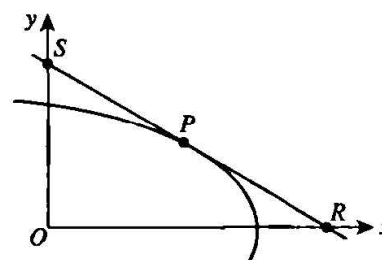


(i) Show that the equation of the tangent to the hyperbola at Q is $x + q^2y = 2cq$. 2

(ii) Show that T has coordinate $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ 2

(iii) If P and Q move so that $pq = k$ where k is a constant, show that the locus of T is a straight line and give its equation in terms of k . 2

- (d) In the diagram, the tangent to the ellipse at $P(\sqrt{2} \cos \theta, 2\sqrt{2} \sin \theta)$ intersect the x -axis at R and the y -axis at S .



(i) Show that the area of $\triangle ORS$ is $\frac{4}{\sin 2\theta}$, 1
where O is the origin. [You may assume the equation of the tangent at P to be

$$\frac{x \cos \theta}{\sqrt{2}} + \frac{y \sin \theta}{2\sqrt{2}} = 1]$$

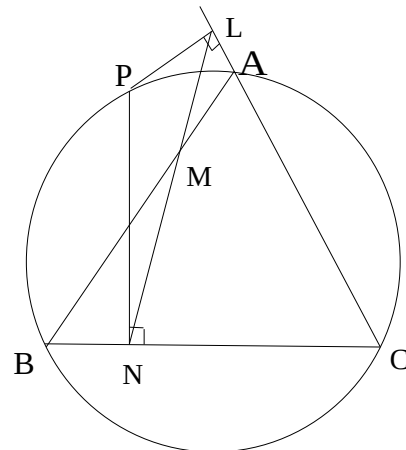
(ii) Find the coordinate of P where the area is a minimum. 2

Question 5

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Marks

- (a) ABC is an acute-angled triangle inscribed in a circle. P is a point on the minor arc AB of the circle. PL and PN are the perpendiculars from P to CA (produced) and CB respectively. LN cuts AB at M.



- (i) Copy the diagram. 1
- (ii) Explain why PNCL is a cyclic quadrilateral. 2
- (iii) Show that $\angle PBM = \angle PNM$. 1
- (iv) Hence deduce that 2
- (α) PBNM is a cyclic quadrilateral 1
- (β) PM is perpendicular to AB. 2

- (b) The roots of $x^3 + px^2 + qx + r = 0$ form an arithmetic progression, prove that $2p^3 + 27r - 9pq = 0$. 3

- (c) Use the following identity to answer the following question.

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

- (i) Solve the equation $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$. 3
- (ii) Hence show that

(α) $\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + 4\sqrt{3} = \tan \frac{5\pi}{24} + \tan \frac{11\pi}{24}$ 2

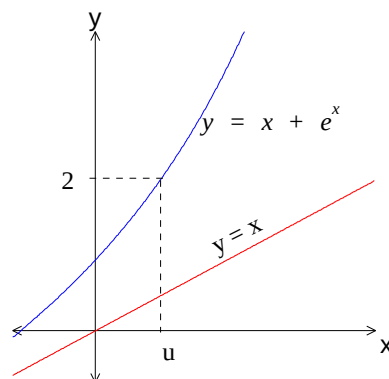
(β) $\tan \frac{5\pi}{24} \tan \frac{7\pi}{24} = \cot \frac{\pi}{24} \cot \frac{11\pi}{24}$ 1

Question 6

(Begin a new page)

Marks

- (a) The diagram shows the graph of the curve $y = x + e^x$, $x \geq 0$. The region bounded by the curve and the line $y = x$ between $x = 0$ and $x = u$ is rotated through one complete revolution about the y -axis.

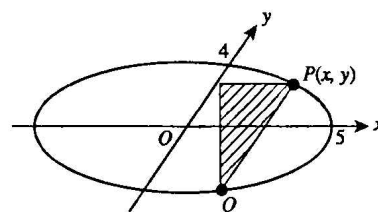


- (i) use the method of cylindrical shells to show that the volume of the solid of revolution is given

$$\text{by } V = 2\pi \int_0^u x e^x dx$$

- (ii) Hence show that $V = 2\pi(3u - u^2 - 1)$

- (b) The base of a certain solid is the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Every cross-section perpendicular to the x -axis is an equilateral triangle. The shaded cross-section is an equilateral triangle with base PQ .



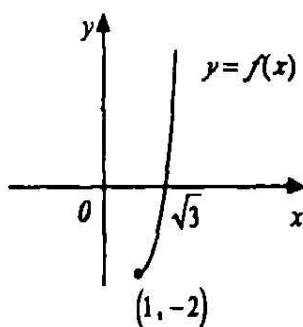
- (i) Show that the shaded cross-sectional area is given by

$$A = \sqrt{3}y^2$$

- (ii) Hence express the cross-sectional area as a function of x .

- (iii) Find the volume of the solid.

- (c) The diagram shows the graph of $y = f(x)$ where $f(x) = x^3 - 3x$; $x \geq 1$



- (i) Copy the diagram. On your diagram sketch the graph of the inverse function $y = f^{-1}(x)$, showing any intercepts on the coordinate axes and the coordinates of any end points. Draw in the line of $y = x$.

- (ii) Find the coordinate of any point of intersection of the curve $y = f(x)$ and $y = f^{-1}(x)$.

- (iii) Hence find the area of the region in the first quadrant bounded by the curve $y = f(x)$, $y = f^{-1}(x)$ and the coordinate axes.

Question 7**(Begin a new page)****Marks**

(a) Solve for x , $\tan^{-1}(4x) - \tan^{-1}(3x) = \tan^{-1}\left(\frac{1}{7}\right)$

3

(b) A sequence of number u_n is defined by

$$u_n = 8u_{n-1} - 15u_{n-2} \quad n \geq 3$$

$$\text{and} \quad u_1 = 2, \quad u_2 = 16$$

(i) Prove that $u_n = 5^n - 3^n$ for $n \geq 1$ by the method of mathematical induction.

4

(ii) Hence show that

2

$$u_1 + u_2 + \dots + u_n = \frac{5^{n+1} - 2 \times 3^{n+1} + 1}{4}$$

(c) If $I_n = \int_0^1 x(1-x)^n dx \quad n = 0, 1, 2, \dots$

(i) Show that $I_n = \frac{n}{n+2} I_{n-1} \quad n = 1, 2, \dots$

2

(ii) Hence deduce that $I_n = \frac{1}{2^{n+2} C_2} \quad n = 1, 2, 3, \dots$

2

(iii) Find n for which $I_n \leq \frac{1}{132}$

2

Question 8**(Begin a new page)****Marks**

(a) Consider the polynomial $P(x) = \frac{x^4}{4} - \frac{x^3}{3} + x^2 - 2x + p$, p is a real number.

(i) Show that $P(x)$ has exactly one turning point for all real values of p .

2

(ii) For what value of p does the equation $P(x) = 0$ have no real roots?

2

(b) A particle moves in a straight line with velocity v and its acceleration given by $a = -v\sqrt{1-v^2}$. The displacement x , of the particle from a fixed origin O is initially zero and its velocity at that time is V .

(i) Show that

4

$$x = \sin^{-1}(V\sqrt{1-v^2} - v\sqrt{1-V^2})$$

(ii) The time that has passed since the particle began its movement is given by t .

3

By considering $a = \frac{dv}{dt}$, show that

$$t = \ln \left[\frac{V(1 + \sqrt{1-v^2})}{v(1 + \sqrt{1-V^2})} \right]$$

[You may assume the following result

$$\int \frac{dv}{v\sqrt{1-v^2}} = \log_e \left(\frac{v}{1 + \sqrt{1-v^2}} \right) \quad , \text{ DON'T PROVE IT.}]$$

(c) Consider $f(x) = \ln(1+x) - \frac{x}{1+x}$ and $g(x) = \ln(1+x) - x$, $x > 0$

(i) Show that $g(x)$ is a decreasing function for all positive values of x .

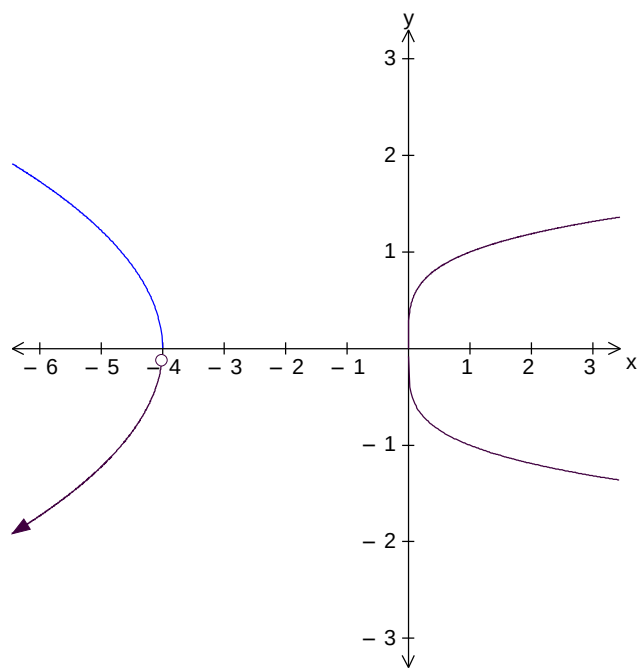
1

(ii) Deduce that $x > \ln(1+x)$

1

(iii) Hence show that $\frac{x}{1+x} < \ln(1+x) < x$

2**End of Paper**



Question 1

$$a) \int_0^1 \frac{e^x}{1+e^x} dx = [\log_e(1+e^x)]_0^1 \\ = \log_e \frac{1+e}{2}$$

$$b) \int \frac{1-\sin x}{\cos^2 x} dx = \int (\sec^2 x - \tan x \sec x) dx \\ = \tan x - \sec x + C$$

$$c) (i) u = 1-x^2 \quad \therefore x^2 = 1-u \\ du = -2x dx \quad x dx = -\frac{du}{2}$$

$$\text{when } x = \frac{\sqrt{3}}{2}, \quad u = \frac{1}{4}$$

$$\text{when } x = 0 \quad u = 1$$

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{x^3}{\sqrt{1-x^2}} dx = \int_1^{\frac{1}{4}} \frac{x^2}{\sqrt{1-x^2}} x dx$$

$$= - \int_1^{\frac{1}{4}} \frac{1-u}{2\sqrt{u}} du$$

$$= \frac{1}{2} \int_{\frac{1}{4}}^1 \left(\frac{1}{\sqrt{u}} - \sqrt{u} \right) du$$

$$= \frac{1}{2} \left[2\sqrt{u} - \frac{2}{3} u^{\frac{3}{2}} \right]_{\frac{1}{4}}^1$$

$$= \frac{1}{2} \left(2 - \frac{2}{3} \right) - \frac{1}{2} \left(2 \cdot \frac{1}{2} - \frac{2}{3} \cdot \frac{1}{8} \right)$$

$$= \frac{5}{24}$$

$$(ii) \int_0^{\frac{\sqrt{3}}{2}} 3x^2 \cos^{-1} x dx = \int_0^{\frac{\sqrt{3}}{2}} \cos^{-1} x d(x^3) \\ = [x^3 \cos^{-1} x]_0^{\frac{\sqrt{3}}{2}} - \int_0^{\frac{\sqrt{3}}{2}} x^3 d \cos^{-1} x \\ = \frac{3\sqrt{3}}{8} \cdot \frac{\pi}{6} + \int_0^{\frac{\sqrt{3}}{2}} \frac{x^3}{\sqrt{1-x^2}} dx \\ = \frac{\sqrt{3}\pi}{16} + \frac{5}{24}$$

$$(d) \text{ Put } t = \tan \frac{x}{2}$$

$$dx = \frac{2 dt}{1+t^2}$$

$$x = \frac{\pi}{3}, \quad t = \frac{1}{\sqrt{3}}$$

$$x = 0 \quad t = 0$$

$$\int_0^{\frac{\pi}{3}} \frac{1}{5-4\cos x} dx = \int_0^{\frac{1}{\sqrt{3}}} \frac{2 dt}{(1+t^2)(5-\frac{4(1-t^2)}{1+t^2})}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2 dt}{5+5t^2-4+4t^2}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2 dt}{1+9t^2}$$

$$= \frac{1}{9} \int_0^{\frac{1}{\sqrt{3}}} \frac{2 dt}{\frac{1}{9} + t^2}$$

$$= \frac{2}{9} \cdot 3 [\tan^{-1} 3t]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{\pi}{9}$$

Q1 (cont'd)

1(e) (i) $(Ax+1)(x+1) + B(x^2+1) = x^2 + 3x$

Put $x=0$ $B+1 = 0$

$\therefore B = -1$

Equating coeff of x^2 $A+B = 1$

$\therefore A = 1-B$

$= 2$

(ii) $\int \frac{x^2+3x}{(x^2+1)(x+1)} dx = \int \frac{2x+1}{x^2+1} - \frac{1}{x+1} dx$

$= \log_e(x^2+1) - \log_e(x+1) + \tan^{-1}x + C$

$= \log_e \frac{x^2+1}{x+1} + \tan^{-1}x + C$

Question 2

(a) $(1+i)^3 = 1+3i-3-i$

$= 2i-2$

$= 2(i-1)$

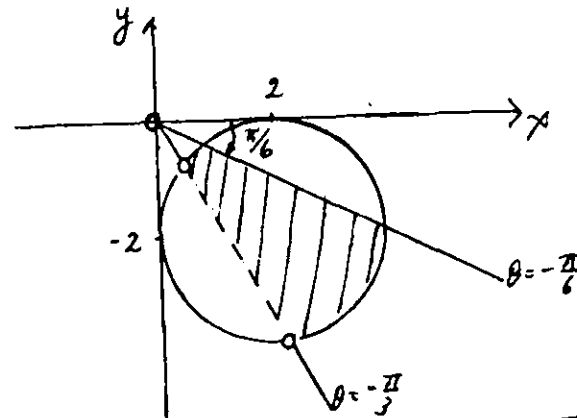
(b) $z = \frac{6-2i}{3+4i} - \frac{6}{5i}$

$= \frac{(6-2i)(3-4i)}{(3+4i)(3-4i)} + \frac{6i}{5}$

$= \frac{18-8-30i}{9+16} + \frac{6i}{5}$

$= \frac{2}{5}$ which is real

(c)



✓ for circle
✓ for $\theta = -\frac{\pi}{6}$
✓ for $\theta = -\frac{\pi}{3}$
(dotted)

d) (i) Let C be the centre of the circle

$|z-3-4i| = 2$

$\therefore C$ is $(3, 4)$.

Let P be the point such that O, C, P are collinear.

$OP = OC + CP$
 $= \sqrt{3^2+4^2} + 2$
 $= 7$

$\therefore$ max value of $|z|$ is 7

(ii) $OC = 5, CQ = 2$

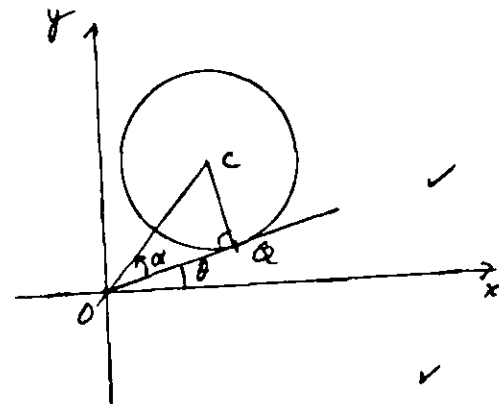
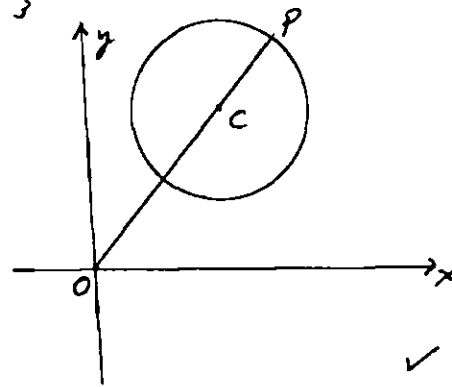
$\therefore \alpha = \sin^{-1} \frac{2}{5}$

but $\alpha + \theta = \arg(3+4i)$
 $= \tan^{-1} \frac{4}{3}$

$\therefore \theta = \tan^{-1} \frac{4}{3} - \sin^{-1} \frac{2}{5}$

or $= 0.5157$ radians

$\therefore$ min $\arg z = 0.5157$ radians.
or $\tan^{-1} \frac{4}{3} - \sin^{-1} \frac{2}{5}$



Q2 (cont'd)

2(e)

(i) Let $z = r \operatorname{cis} \theta$,
then $\bar{z} = r \operatorname{cis}(-\theta)$

$$\therefore \frac{z}{\bar{z}} = \frac{r \operatorname{cis} \theta}{r \operatorname{cis}(-\theta)}$$

$$= \operatorname{cis} 2\theta$$

$$\therefore \arg\left(\frac{z}{\bar{z}}\right) = 2 \arg z.$$

(ii) from (i) $\arg\left(\frac{z}{\bar{z}}\right) = 2 \arg z$

$$\arg\left(\frac{z-1}{\bar{z}-1}\right) = 2 \arg(z-1)$$

Since given $\frac{z}{\bar{z}} = -\frac{z-1}{\bar{z}-1}$

$$\therefore \arg\left(\frac{z}{\bar{z}}\right) = \arg\left(-\frac{z-1}{\bar{z}-1}\right)$$

$$= \arg(-1) + \arg\left(\frac{z-1}{\bar{z}-1}\right)$$

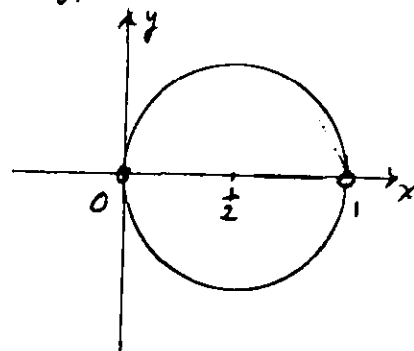
$$2 \arg z = \pi + 2 \arg(z-1) \text{ or } -\pi + 2 \arg(z-1)$$

$$\therefore \arg z = \arg(z-1) + \frac{\pi}{2} \text{ or } \arg z = \arg(z-1) - \frac{\pi}{2}$$

(iii) From (ii) $\arg z - \arg(z-1) = \pm \frac{\pi}{2}$

$$\arg\left(\frac{z}{z-1}\right) = \pm \frac{\pi}{2}$$

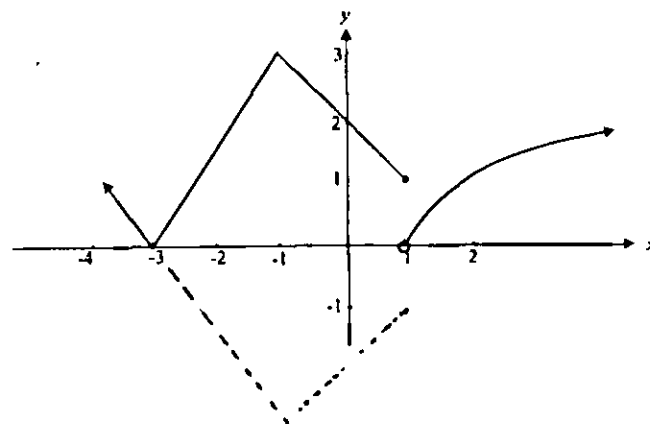
$\therefore$ Locus of $P(z)$ is



✓ 2 semi-circles
✓ excluding $z=0$
and $z=1$.

Question 3

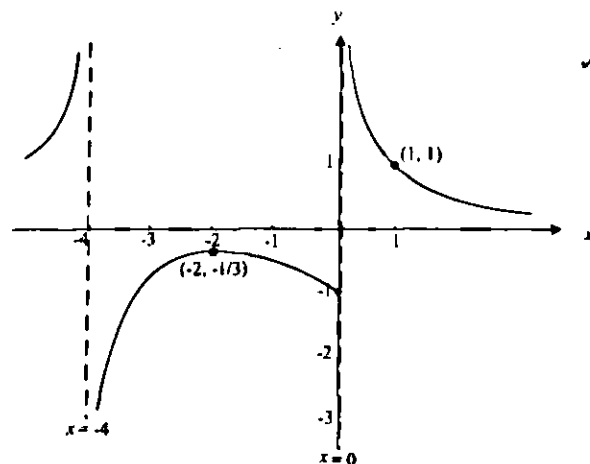
The graph of $y = |f(x-1)|$ is shown below.



✓ for
 $y = f(x-1)$

✓ for $|f(x-1)|$

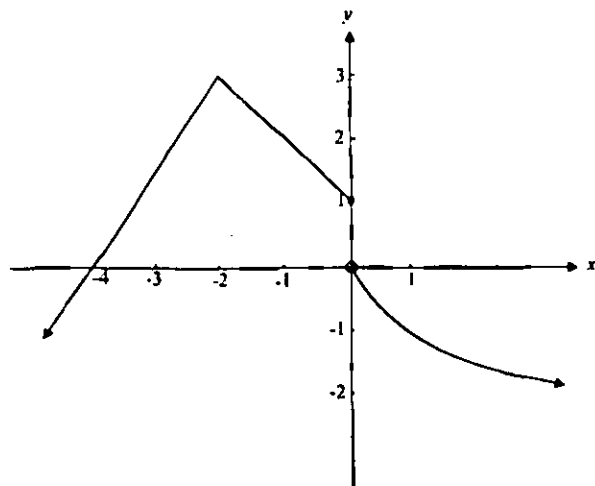
(ii) The graph of $y = \frac{1}{f(x)}$ is shown below.



✓ for asymptotes

✓ for correct
end points.

(iii) The graph of $y = -f(x)$ is shown below.



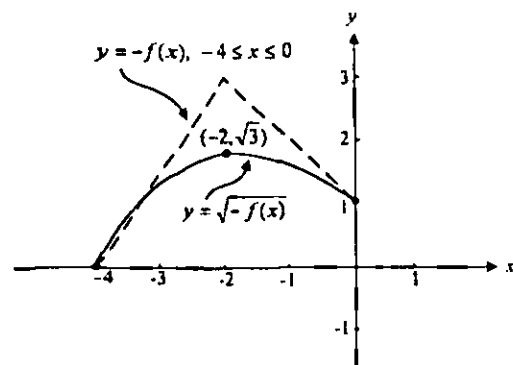
Only those values in the domain of $y = -f(x)$ for which $y \geq 0$ are included in the graph of $y = \sqrt{-f(x)}$. That is, only values of x where $-4 \leq x \leq 0$ are included.

For values of $y = -f(x)$ that are less than 1, $\sqrt{-f(x)} > -f(x)$.

For values of $y = -f(x)$ that are greater than 1, $\sqrt{-f(x)} < -f(x)$.

Note that $\sqrt{-f(0)} = 1$ and $\sqrt{-f(-4)} = 0$.

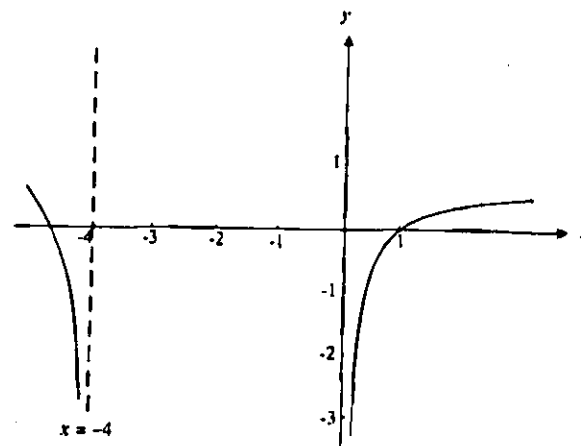
The graph of $y = -f(x)$ for $-4 \leq x \leq 0$ together with the graph of $y = \sqrt{-f(x)}$ are shown below.



- ✓ for correct shape
- ✓ for correct curvature above & below $y = 1$

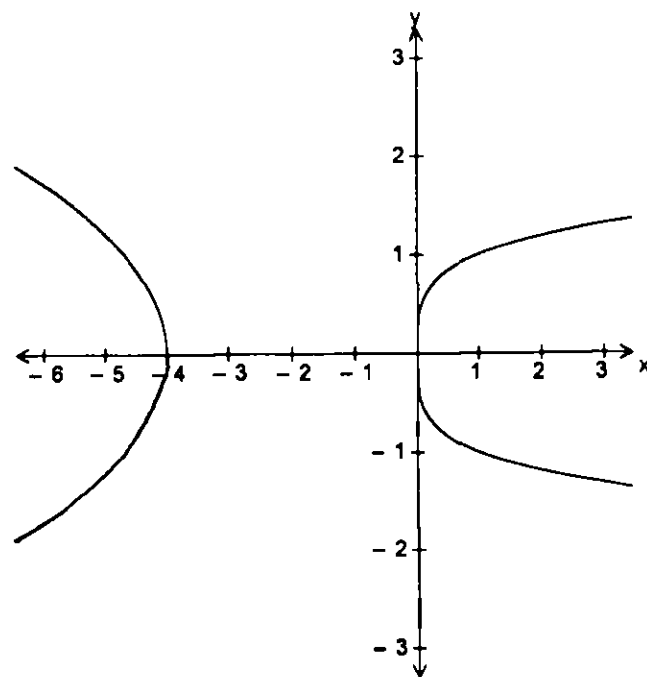
(iv) The graph of $y = \ln(f(x))$ only exists for $f(x) > 0$, that is, for $x < -4$ and $x > 0$.

The graph of $y = \ln(f(x))$ is shown below.



- ✓ for asymptote
- ✓ for correct shape

(v) $y^2 = f(x)$



- ✓ for $y = \sqrt{f(x)}$
- ✓ for the other half

Q3(b)

(i) Since complex roots of any polynomial equation with real coefficients occur in conjugate pairs.

$\therefore$ if $z = 3-i$ is a root, then $\bar{z} = 3+i$ must also be a root.

(ii) Let the other root be β .

$\therefore$ Roots are $3-i, 3+i, \beta$.

Sum of roots $(3-i) + (3+i) + \beta = 0$

$$6 + \beta = 0$$

$$\therefore \beta = -6$$

$\therefore$ Roots are $3-i, 3+i, -6$.

(iii) Since $P(-6) = 0$

$$\therefore (-6)^3 + k(-6) + 60 = 0$$

$$-216 - 6k + 60 = 0$$

$$\therefore k = -26$$

(c)

$$2z^3 = 9 + 3\sqrt{3}i$$

$$= 6\sqrt{3} \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= 6\sqrt{3} \operatorname{cis} \frac{\pi}{6}$$

$$z^3 = 3\sqrt{3} \operatorname{cis} \frac{\pi}{6}$$

$$\therefore z = \sqrt{3} \operatorname{cis} \left(\frac{\frac{\pi}{6} + 2k\pi}{3} \right) \quad k = 0, 1, 2$$

$$\therefore r = \sqrt{3}$$

$$\text{smallest value of } \theta = \frac{\pi}{18}$$

Question 4

(a) (i) Gradient of the line perpendicular to the asymptote is $-\frac{a}{b}$

$\therefore$ Equation of the line is

$$y - 0 = -\frac{a}{b}(x - ae)$$

$$y = -\frac{ax}{b} + \frac{a^2e}{b}$$

$$by = -ax + a^2e$$

$$\text{ie } ax + by - a^2e = 0$$

(1) ✓

(ii) when (1) meets $y = \frac{bx}{a}$

$$ax + b\left(\frac{bx}{a}\right) - a^2e = 0$$

$$(a^2 + b^2)x - a^3e = 0$$

$$x = \frac{a^3e}{a^2 + b^2}$$

$$= \frac{a^3e}{a^2 + a^2e^2 - a^2}$$

$$\therefore b^2 = a^2(e^2 - 1)$$

$$= \frac{a}{e}$$

$\therefore$ They meet at the point where $x = \frac{a}{e}$ which lies on the corresponding directrix.

$$Q4(b) \quad x^4 + 3xy - 2y^2 + 13 = 0$$

$$4x^3 + 3y + 3xy' - 4y y' = 0$$

$$\therefore (3x - 4y)y' = -(4x^3 + 3y)$$

$$y' = -\frac{4x^3 + 3y}{3x - 4y}$$

$$\therefore \text{Gradient at } (-1, 2) = -\frac{4(-1)^3 + 3(2)}{3(-1) - 4(2)}$$

$$= \frac{2}{11}$$

$$\therefore \text{Gradient of normal} = -\frac{11}{2}$$

$\therefore$ Equation of normal is

$$y - 2 = -\frac{11}{2}(x + 1)$$

$$2y - 4 = -11x - 11$$

$$11x + 2y + 7 = 0$$

$$(c) (i) \quad xy = c^2$$

$$\therefore y = \frac{c^2}{x}$$

$$y' = -\frac{c^2}{x^2}$$

$$\therefore \text{at } Q \quad y' = -\frac{c^2}{c^2 q^2}$$

$$= -\frac{1}{q^2}$$

$\therefore$ Equation of the tangent at Q is

$$y - \frac{c}{q} = -\frac{1}{q^2}(x - cq)$$

$$q^2 y - cq = -x + cq$$

$$\text{i.e. } x + q^2 y = 2cq \quad (1)$$

(ii) Similarly, tangent at P is

$$x + p^2 y = 2cp \quad (2)$$

$$(1) - (2) \quad (q^2 - p^2)y = 2c(q - p)$$

$$\therefore y = \frac{2c(q - p)}{q^2 - p^2}$$

$$= \frac{2c}{p + q}$$

$$\text{Put into (1)} \quad x + \frac{2cq^2}{p + q} = 2cq$$

$$\therefore x = \frac{2cpq}{p + q}$$

$$\therefore T \text{ is } \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$$

$$(iii) \text{ At } T \quad x = \frac{2cpq}{p + q} \quad (3)$$

$$y = \frac{2c}{p + q} \quad (4)$$

4(c) (cont'd)

$$\begin{aligned} \frac{x}{y} &= pq \\ &= k \quad (\text{given}) \end{aligned}$$

$$\therefore x = ky$$

$\therefore$ Locus of T is the straight line $x = ky$

$$\text{or } y = \frac{x}{k}$$

(d) Equation of RS:

$$\frac{x \cos \theta}{\sqrt{2}} + \frac{y \sin \theta}{2\sqrt{2}} = 1$$

$$\therefore \text{At R } y = 0.$$

$$\therefore x = \frac{\sqrt{2}}{\cos \theta} \quad R\left(\frac{\sqrt{2}}{\cos \theta}, 0\right)$$

$$\text{At S, } x = 0$$

$$y = \frac{2\sqrt{2}}{\sin \theta} \quad S\left(0, \frac{2\sqrt{2}}{\sin \theta}\right)$$

$$\therefore \text{Area of } \triangle ORS = \frac{1}{2} OR \times OS$$

$$= \frac{1}{2} \cdot \frac{\sqrt{2}}{\cos \theta} \cdot \frac{2\sqrt{2}}{\sin \theta}$$

$$= \frac{4}{2 \sin \theta \cos \theta}$$

$$= \frac{4}{\sin 2\theta}$$

$$\begin{aligned} \text{(ii)} \quad \triangle ORS \text{ is minimum when } \sin 2\theta &= 1 \therefore \theta = \frac{\pi}{4} \\ \text{ie } \sin \theta &= \cos \theta = \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \therefore P \text{ is } \left(\sqrt{2} \cdot \frac{1}{\sqrt{2}}, 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}\right) \\ = (1, 2). \end{aligned}$$

Question 5

$$\begin{aligned} \text{(ii)} \quad \angle PNC + \angle PLC &= 90^\circ + 90^\circ \quad (\text{given}) \\ &= 2 \text{ rt } \angle s \end{aligned}$$

$\therefore PNC L$ is a cyclic quad (opp $\angle$'s of cyclic quad are supplementary)

$$\text{(iii)} \quad \angle PBM = \angle PBA$$

$$= \angle PCA$$

($\angle$'s in same segment)

$$= \angle PCL$$

($\angle$'s in same segment
(PNCL is a cyclic quad))

$$= \angle PNL$$

$$= \angle PNM$$

$$\text{(iv)} \quad (\alpha) \quad \angle PBM = \angle PNM \quad (\text{proved})$$

$\therefore PBNM$ is a cyclic quad
($\angle$'s subtended by same arc in same segment are equal)

$$\text{(}\beta\text{)} \quad \angle PMB = \angle PNB$$

($\angle$'s in same segment are equal,
(PBNM is a cyclic quad))

$$= 90^\circ$$

($\angle PNB = 90^\circ$, given)

$$\therefore PM \perp AB$$

Q 5(b)

Let the roots be $\alpha-d, \alpha, \alpha+d$.

$\therefore$ sum of roots: $\alpha-d+\alpha+\alpha+d = -p$

$$\therefore p = -3\alpha$$

Sum of roots taken 2 at a time:

$$\alpha(\alpha-d) + \alpha(\alpha+d) + (\alpha-d)(\alpha+d) = q$$

$$3\alpha^2 - d^2 = q$$

$$q = 3\alpha^2 - d^2$$

Product of roots

$$\alpha(\alpha-d)(\alpha+d) = -r$$

$$\alpha^3 - \alpha d^2 = -r$$

$$r = \alpha d^2 - \alpha^3$$

$$\therefore 2p^3 + 27r - 9pq$$

$$= 2(-3\alpha)^3 + 27(\alpha d^2 - \alpha^3) - 9(-3\alpha)(3\alpha^2 - d^2)$$

$$= -54\alpha^3 + 27\alpha d^2 - 27\alpha^3 + 81\alpha^3 - 27\alpha d^2$$

$$= 0$$

(c) (i) $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$

$$\therefore 1 - 6x^2 + x^4 = 4\sqrt{3}x - 4\sqrt{3}x^3$$

$$\therefore \frac{(4x - 4x^3)\sqrt{3}}{1 - 6x^2 + x^4} = 1$$

$$\frac{4x - 4x^3}{1 - 6x^2 + x^4} = \frac{1}{\sqrt{3}}$$

Let $x = \tan \theta$

$$\therefore \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta} = \frac{1}{\sqrt{3}}$$

$$\tan 4\theta = \frac{1}{\sqrt{3}}$$

$$\therefore 4\theta = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6}, \frac{25\pi}{6} \dots$$

$$\theta = \frac{\pi}{24}, \frac{7\pi}{24}, \frac{13\pi}{24}, \frac{19\pi}{24}, \frac{25\pi}{24} \dots$$

$\therefore$ solutions are $x = \tan \frac{\pi}{24}, \tan \frac{7\pi}{24}, \tan \frac{13\pi}{24}, \tan \frac{19\pi}{24}$

[since $\tan \frac{25\pi}{24} = \tan \frac{\pi}{24}$ etc...]

(ii) (a) Sum of roots:

$$\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + \tan \frac{13\pi}{24} + \tan \frac{19\pi}{24} = -4\sqrt{3}$$

$$\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + \tan(\pi - \frac{11\pi}{24}) + \tan(\pi - \frac{5\pi}{24}) = -4\sqrt{3}$$

$$\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} - \tan \frac{11\pi}{24} - \tan \frac{5\pi}{24} = -4\sqrt{3}$$

$$\therefore \tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + 4\sqrt{3} = \tan \frac{5\pi}{24} + \tan \frac{11\pi}{24}$$

(b) Product of roots

$$\tan \frac{\pi}{24} \cdot \tan \frac{7\pi}{24} (-\tan \frac{5\pi}{24}) (-\tan \frac{11\pi}{24}) = 1$$

$$\text{ie } \tan \frac{5\pi}{24} \tan \frac{\pi}{24} = \frac{1}{\tan \frac{\pi}{24} \tan \frac{11\pi}{24}}$$

$$= \cot \frac{\pi}{24} \cot \frac{11\pi}{24}$$

Question 6

(i) $r = x$
 $h = (x + e^x) - x$
 $= e^x$ ✓

∴ Volume of the cylindrical shell

$$\delta V = 2\pi r h \delta x$$

$$= 2\pi x e^x \delta x$$
 ✓

$$\therefore \text{Volume} = \int dV$$

$$= \int_0^u 2\pi x e^x dx$$

$$= 2\pi \int_0^u x e^x dx$$
 ✓

(ii) $V = 2\pi \int_0^u x e^x dx$

$$= 2\pi [x e^x]_0^u - 2\pi \int_0^u e^x dx$$

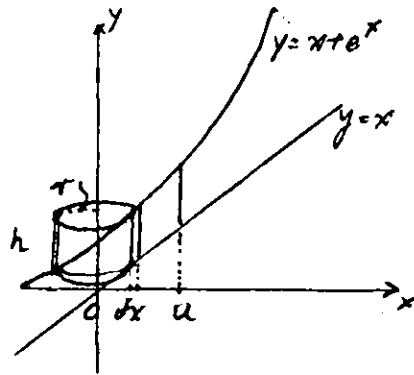
$$= 2\pi u e^u - 2\pi [e^x]_0^u$$

$$= 2\pi u e^u - 2\pi e^u + 2\pi$$
 ✓
$$= 2\pi (u e^u - e^u + 1)$$
 (1)

Since $y = 2$ when $x = u$ on $y = x + e^x$

$$\therefore u + e^u = 2$$

$$e^u = 2 - u$$
 (2) ✓



∴ Put (2) into (1)

$$V = 2\pi [u(2-u) - (2-u) + 1]$$

$$= 2\pi (2u - u^2 - 2 + u + 1)$$

$$= 2\pi (3u - u^2 - 1)$$
 ✓

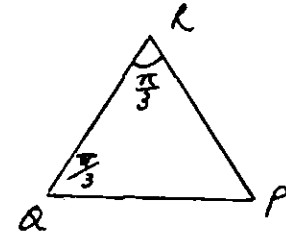
(b) (i) $PQ = 2y$

$$\therefore QR = PQ = PR = 2y$$

$$\therefore \Delta PQR = \frac{1}{2} PQ \cdot QR \sin \frac{\pi}{3}$$

$$= \frac{1}{2} \cdot 2y \cdot 2y \cdot \frac{\sqrt{3}}{2}$$

$$= \sqrt{3} y^2$$
 ✓



(ii) $\Delta PQR = \sqrt{3} \cdot 16 (1 - \frac{x^2}{25})$

$$= \frac{16\sqrt{3}(25-x^2)}{25}$$
 ✓

(iii) Volume of the slice

$$\delta V = \frac{16\sqrt{3}(25-x^2)}{25} \delta x$$

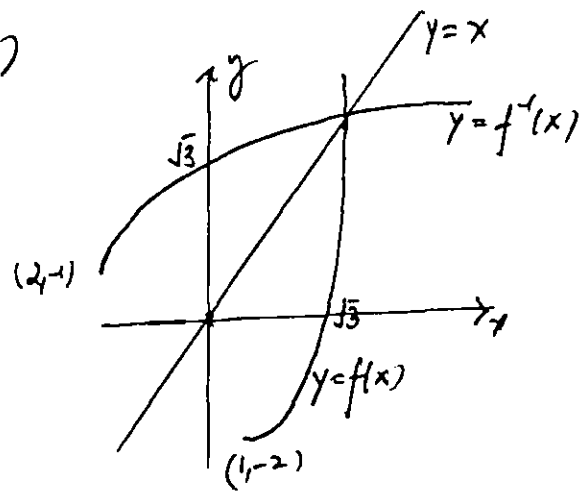
$$\therefore V = \frac{16\sqrt{3}}{25} \int_{-5}^5 (25-x^2) dx$$

$$= \frac{32\sqrt{3}}{25} \int_0^5 (25-x^2) dx$$
 ✓
$$= \frac{32\sqrt{3}}{25} \left[25x - \frac{x^3}{3} \right]_0^5$$

$$= \frac{320\sqrt{3}}{3} \text{ unit}^3$$
 ✓

Q6 (c)

(i)



✓ for end-pt
& intercept
✓ for correct
shape

(ii) At pt of intersection,

$$x = f(x)$$

$$x = x^3 - 3x$$

$$\therefore x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x(x-2)(x+2)$$

$$\therefore x = 0, x = 2, x = -2$$

but domain of $y = f(x)$ is $x \geq 1$

$$\therefore x = 2$$

$\therefore$ pt of intersection is (2,2)

$$(iii) \text{ Area} = 2 \left[\frac{1}{2} \times 2 \times 2 - \int_1^2 x^3 - 3x dx \right]$$

$$= 2 \left[2 - \left[\frac{x^4}{4} - \frac{3x^2}{2} \right]_1^2 \right]$$

$$= \frac{7}{2} \text{ unit}^2$$

Question 7

$$(a) \quad \tan^{-1}(4x) - \tan^{-1}(3x) = \tan^{-1}\left(\frac{1}{7}\right)$$

$$\tan\left(\tan^{-1}(4x) - \tan^{-1}(3x)\right) = \frac{1}{7}$$

$$\frac{4x - 3x}{1 + (4x)(3x)} = \frac{1}{7}$$

$$7x = 1 + 12x^2$$

$$\therefore 12x^2 - 7x + 1 = 0$$

$$(3x - 1)(4x - 1) = 0$$

$$\therefore x = \frac{1}{3} \text{ or } x = \frac{1}{4}$$

$$(b) (i) \text{ when } n=1 \quad 5^1 - 3^1 = 2$$

$$\text{when } n=2 \quad 5^2 - 3^2 = 16$$

$$\therefore u_n = 5^n - 3^n \text{ is true for } n=1, 2.$$

Assume it is true for $n=k+1, k-2$.

$$\text{ie } u_{k-1} = 5^{k-1} - 3^{k-1}$$

$$u_{k-2} = 5^{k-2} - 3^{k-2}$$

$$\therefore u_k = 8u_{k-1} - 15u_{k-2}$$

$$= 8(5^{k-1} - 3^{k-1}) - 15(5^{k-2} - 3^{k-2})$$

$$= 40 \cdot 5^{k-2} - 24 \cdot 3^{k-2} - 15 \cdot 5^{k-2} + 15 \cdot 3^{k-2}$$

$$= 25 \cdot 5^{k-2} - 9 \cdot 3^{k-2}$$

$$= 5^k - 3^k$$

$\therefore$ It will be true for $n=k$ if it is true for $n=k-1, n=k-2$ ✓

Since it is proven true for $k=1, 2$, $\therefore$ it will be true for $n=3, 4, 5, \dots$ i.e. true for all positive integers n .

(ii) $u_1 + u_2 + u_3 + \dots + u_n$

$$= (5-3) + (5^2-3^2) + (5^3-3^3) + \dots + (5^n-3^n)$$

$$= (5+5^2+5^3+\dots+5^n) - (3+3^2+3^3+\dots+3^n)$$
 ✓

$$= \frac{5(5^n-1)}{5-1} - \frac{3(3^n-1)}{3-1}$$

$$= \frac{5^{n+1}-5}{4} - \frac{3^{n+1}-3}{2}$$

$$= \frac{5^{n+1}-5-2 \times 3^{n+1}+6}{4}$$

$$= \frac{5^{n+1}-2 \times 3^{n+1}+1}{4}$$
 ✓

(c) (i) $I_n = \int_0^1 x(1-x)^n dx$

$$= \left[\frac{x^2}{2}(1-x)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} d(1-x)^n$$

$$= \frac{n}{2} \int_0^1 x^2(1-x)^{n-1} dx$$

$$2I_n = n \int_0^1 x^2(1-x)^{n-1} dx$$

$$= n \int_0^1 x[1-(1-x)](1-x)^{n-1} dx$$
 ✓

$$= n \int_0^1 x(1-x)^{n-1} dx - n \int_0^1 x(1-x)^n dx$$

$$= n I_{n-1} - n I_n$$

$$(n+2)I_n = n I_{n-1}$$

$$\therefore I_n = \frac{n}{n+2} I_{n-1}$$
 ✓

(ii) $I_0 = \int_0^1 x dx = \frac{1}{2}$

$$I_1 = \frac{1}{3} I_0$$

$$I_2 = \frac{2}{4} I_1$$

$$I_3 = \frac{3}{5} I_2$$

$$\vdots$$

$$I_{n-1} = \frac{n-1}{n+1} I_{n-2}$$

$$I_n = \frac{n}{n+2} I_{n-1}$$

Multiplying

$$I_n = \frac{n(n-1)(n-2)\dots 1}{(n+2)(n+1)\dots 5 \cdot 4 \cdot 3} \cdot \frac{1}{2}$$
 ✓

$$\text{ie } I_n = \frac{n! 2!}{(n+2)! 2}$$

$$= \frac{1}{2^{n+2} C_2}$$

$$(iii) I_n \leq \frac{1}{132}$$

$$\frac{1}{2^{n+2} C_2} \leq \frac{1}{132}$$

$$66 \leq {}^{n+2}C_2$$

$$\text{ie } \frac{(n+2)(n+1)}{2} \geq 66$$

$$n^2 + 3n + 2 \geq 132$$

$$n^2 + 3n - 130 \geq 0$$

$$(n-10)(n+13) \geq 0$$

$$\therefore n \leq -13 \text{ or } n \geq 10$$

$$\therefore n \geq 10$$

Question 8

$$(a) p(x) = \frac{x^4}{4} - \frac{x^3}{3} + x^2 - 2x + p$$

$$(i) p'(x) = x^3 - x^2 + 2x - 2$$

$$= x^2(x-1) + 2(x-1)$$

$$= (x^2+2)(x-1)$$

$$\text{when } p'(x) = 0$$

$$(x^2+2)(x-1) = 0$$

$$x = 1 \quad x^2+2 \neq 0.$$

$\therefore$ There is only one turning point.

(ii) Since $p(x)$ is a quartic polynomial,

$$\& p''(x) = 3x^2 - 2x + 2$$

$$p''(1) = 3 > 0$$

$\therefore$ The turning pt is a minimum turning pt.

The equation $p(x) = 0$ will have no real roots if the graph of $y = p(x)$ does not touch or cut the x -axis, ie when $p(1) > 0$

$$\text{ie } \frac{1}{4} - \frac{1}{3} + 1 - 2 + p > 0$$

$$p > \frac{13}{12}$$

Q.P(b)

$$(i) \quad a = -v\sqrt{1-v^2}$$

$$\therefore v \frac{dv}{dx} = -v\sqrt{1-v^2}$$

$$\int_v^V \frac{dv}{\sqrt{1-v^2}} = - \int_0^x dx$$

$$\left[\sin^{-1} v \right]_v^V = -x$$

$$\therefore x = \sin^{-1} V - \sin^{-1} v$$

$$\text{Let } \alpha = \sin^{-1} V, \quad \beta = \sin^{-1} v \quad \begin{matrix} \sin \alpha = V, & \sin \beta = v \\ \cos \alpha = \sqrt{1-V^2}, & \cos \beta = \sqrt{1-v^2} \end{matrix}$$

$$x = \alpha - \beta$$

$$\sin x = \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \sin \beta \cos \alpha$$

$$= V\sqrt{1-v^2} - v\sqrt{1-V^2}$$

$$\therefore x = \sin^{-1} [V\sqrt{1-v^2} - v\sqrt{1-V^2}]$$

$$(ii) \quad a = \frac{dv}{dt} = -v\sqrt{1-v^2}$$

$$\int_v^V \frac{dv}{v\sqrt{1-v^2}} = - \int_0^t dt$$

$$\therefore \left[\ln v - \ln(1 + \sqrt{1-v^2}) \right]_v^V = -t$$

$$\therefore t = \left[\ln V - \ln(1 + \sqrt{1-V^2}) \right] - \left[\ln v - \ln(1 + \sqrt{1-v^2}) \right]$$

$$= \ln \frac{V}{v} + \ln \frac{1 + \sqrt{1-v^2}}{1 + \sqrt{1-V^2}}$$

$$= \ln \frac{V(1 + \sqrt{1-v^2})}{v(1 + \sqrt{1-V^2})}$$

$$(c) (i) \quad g(x) = \ln(1+x) - x$$

$$g'(x) = \frac{1}{1+x} - 1$$

$$= -\frac{x}{1+x} < 0 \quad \text{since } x > 0$$

$\therefore g(x)$ is a decreasing function for all $x > 0$

$$(ii) \quad \therefore g(x) > g(0) \quad \text{for all } x > 0$$

$$\therefore \ln(1+x) - x > 0$$

$$\ln(1+x) > x \quad \forall x > 0$$

$$(iii) \quad f(x) = \ln(1+x) - \frac{x}{1+x}$$

$$= \ln(1+x) - \left[1 - \frac{1}{1+x} \right]$$

$$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2}$$

$$p(c) \text{ (cont'd)} = \frac{1+x - 1}{(1+x)^2}$$

$$= \frac{x}{(1+x)^2} > 0 \quad \forall x > 0$$

$\therefore f(x)$ is an increasing function of x for all $x > 0$

$$\therefore f(x) > f(0) \quad \text{for all } x > 0 \quad \checkmark$$

$$\ln(1+x) - \frac{x}{1+x} > 0$$

$$\ln(1+x) > \frac{x}{1+x}$$

but from (c)(ii)

$$x > \ln(1+x)$$

$$\therefore \frac{x}{1+x} < \ln(1+x) < x$$

} $\checkmark$